WEST BENGAL STATE UNIVERSITY

B.Sc. Major 1st Semester Examination, 2023-24

CMSDSC101T-COMPUTER SCIENCE (MAJOR)

Time Allotted: 2 Hours | Full Marks: 50

The figures in the margin indicate full marks.
Candidates should answer in their own words and adhere to the word limit as practicable.
All symbols are of usual significance.

1. Answer any *five* questions from the following: 2×5 = 10

(a) What is the function of '&' operator?

(b) What is meant by return type of a function?

(c) What is the use of static variables in C?

(d) What is pointer in C?

(e) What is typedef in C?

(f) State the formula required to calculate the memory location of an element in row major order and column major order.

(g) What are the types of operators?

(h) If $a = 2$ and $b = 5$ then what will be the output of the expression?

$$(++a)+(b--)-(a++)+(++b)$$

Answer any *five* questions from the following 8×5 = 40

2. (a) What are the different loops in C? 4+4

(b) Demonstrate the operational difference of '*while*' and '*do-while*' loops in C.

3. Write the function of following string function — 2+2+4

strlwr (stringname)

strcmp (firststring, secondstring)

Write the differences between macro and function. Write a C program to sort an array using pointer.

4. (a) Write a C program to find the second maximum and second minimum element from an array along with their positions. 6+2

(b) What is the difference between system call and library function?

5. How is source code is different from object code? What is static and dynamic memory allocation? What is the difference between i++ & ++i ? 3+3+2

6. (a) What is command line argument? Explain with a simple program code. 3+3+2

(b) Write the difference between static and dynamic memory allocation.

(c) Explain recursion with a short program code.

7. (a) What are the different methods of passing parameters to a function? 4+4

(b) Demonstrate with a program in C how address of a variable can be passed to a function.

8. (a) Write a program in C to create a pointer variable of integer type, use it to point to an array and display the values using the pointer variable. 5+3

(b) What is a pointer to a pointer? Give example.

9. Can we return multiple values from a C function? Write a C program to check if two matrices are equals or not. 2+6

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